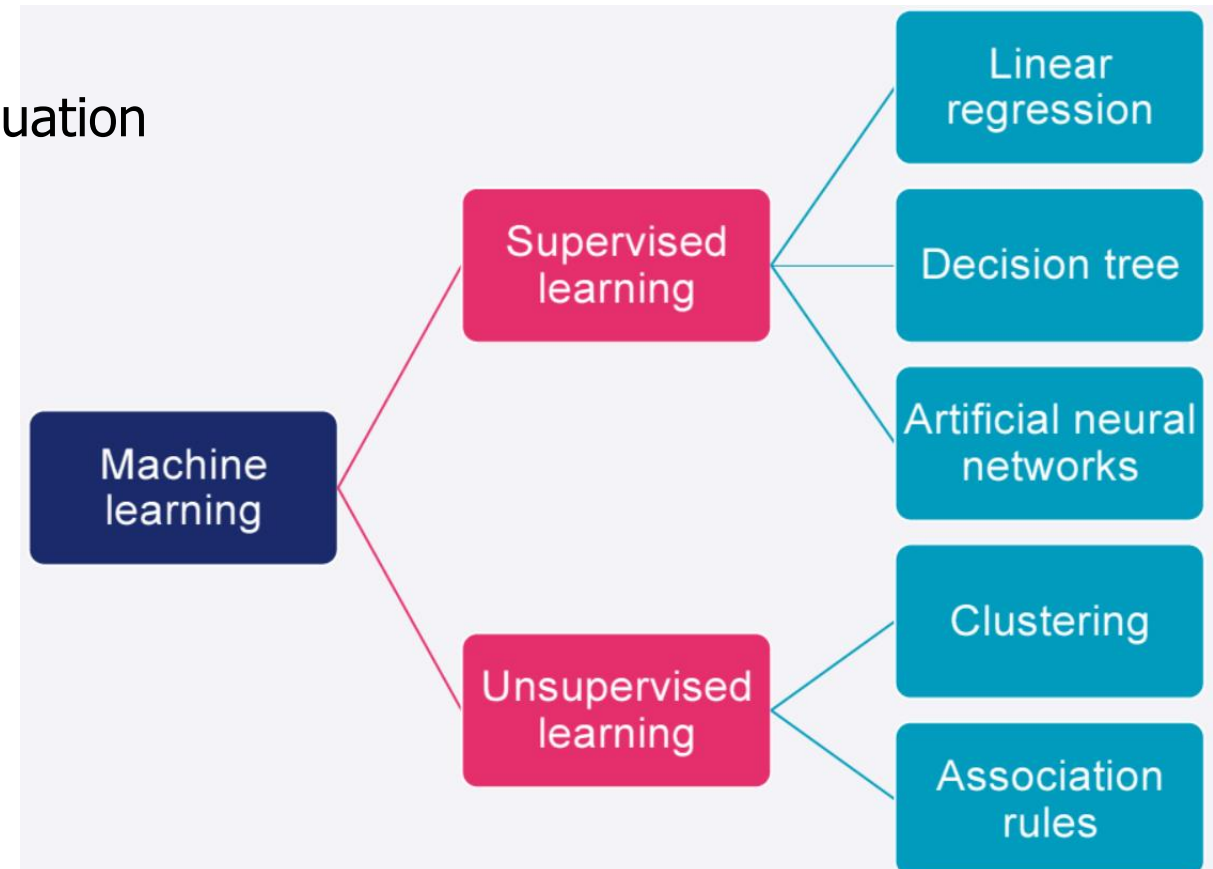


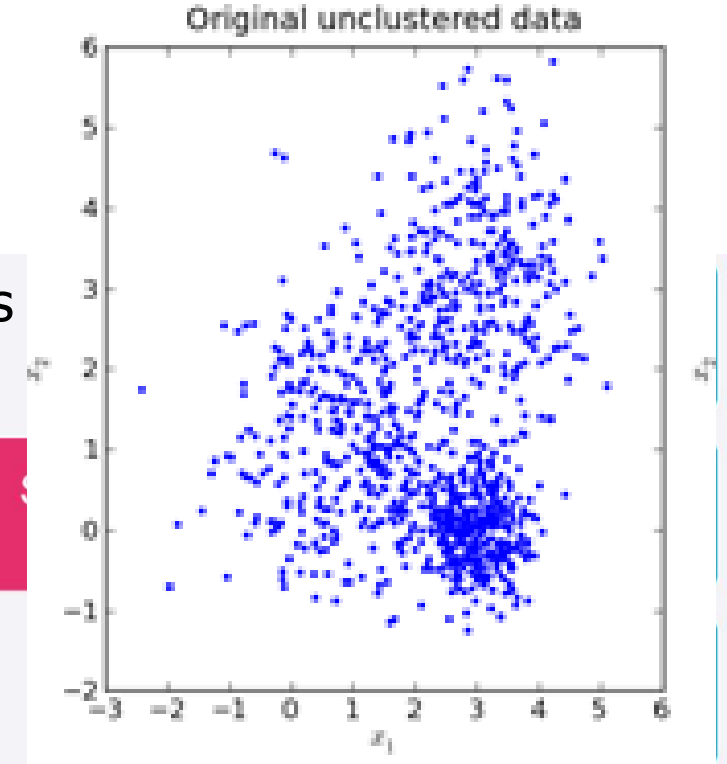
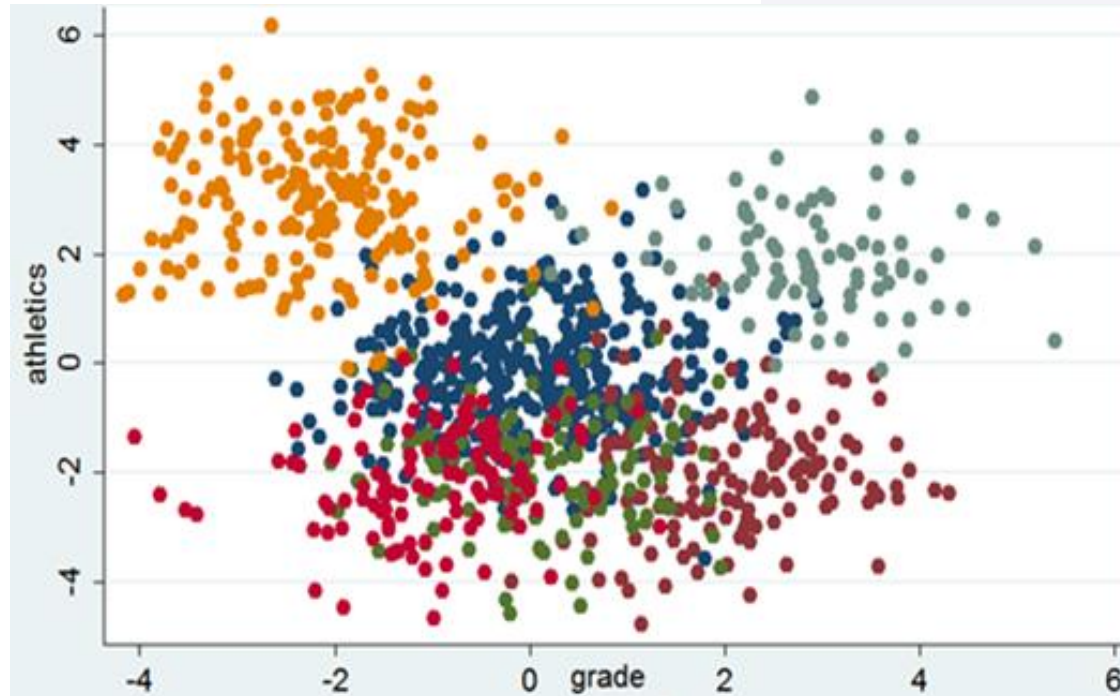
Machine Learning : Un-supervised Learning

- ▶ Discover hidden patterns or structures from **unlabelled** data
 - No need to label data
 - However, no **ground truth** for accurate evaluation
- A lot of real-world applications
 - Marketing: customer segmentation
 - Research: document clustering
 - Finance: fraud detection
 - Science: clustering genes
 - Medical: clustering images



Clustering

- ▶ Group similar objects
 - **Given:** Un-labeled data set D and Similarity/distance metric
 - **Goal:** Find 'natural' partitioning, or groups of similar data points



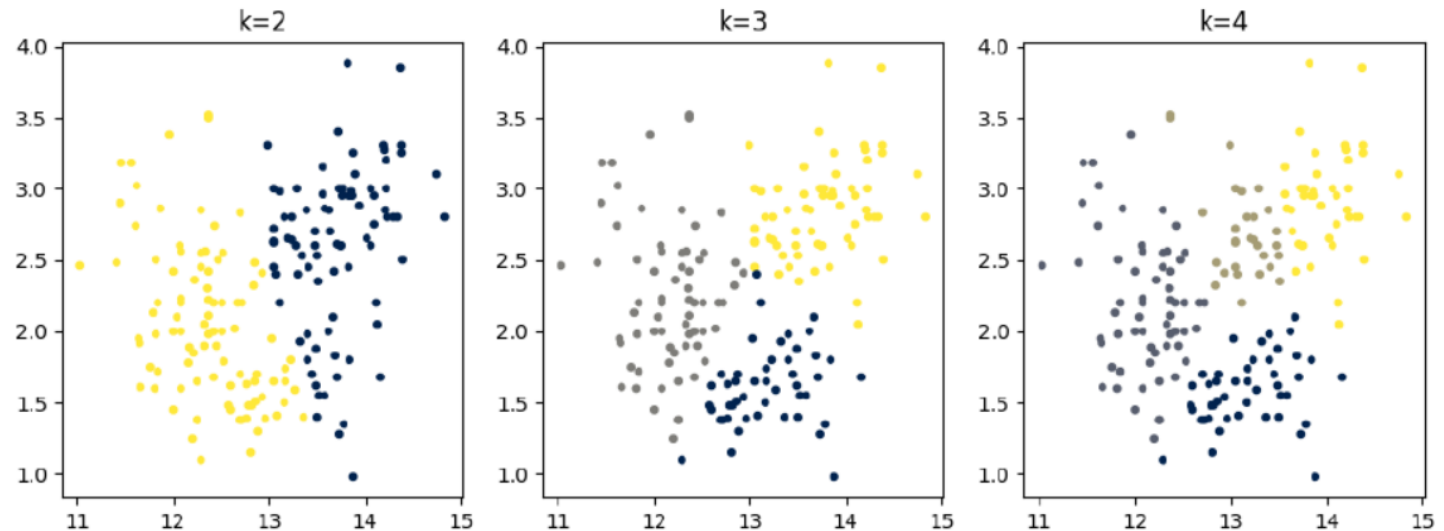
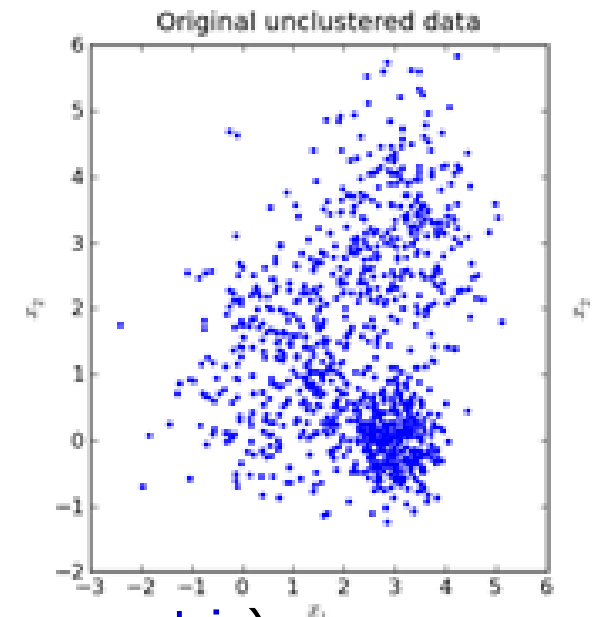
Unsupervised learning

Clustering

Association rules

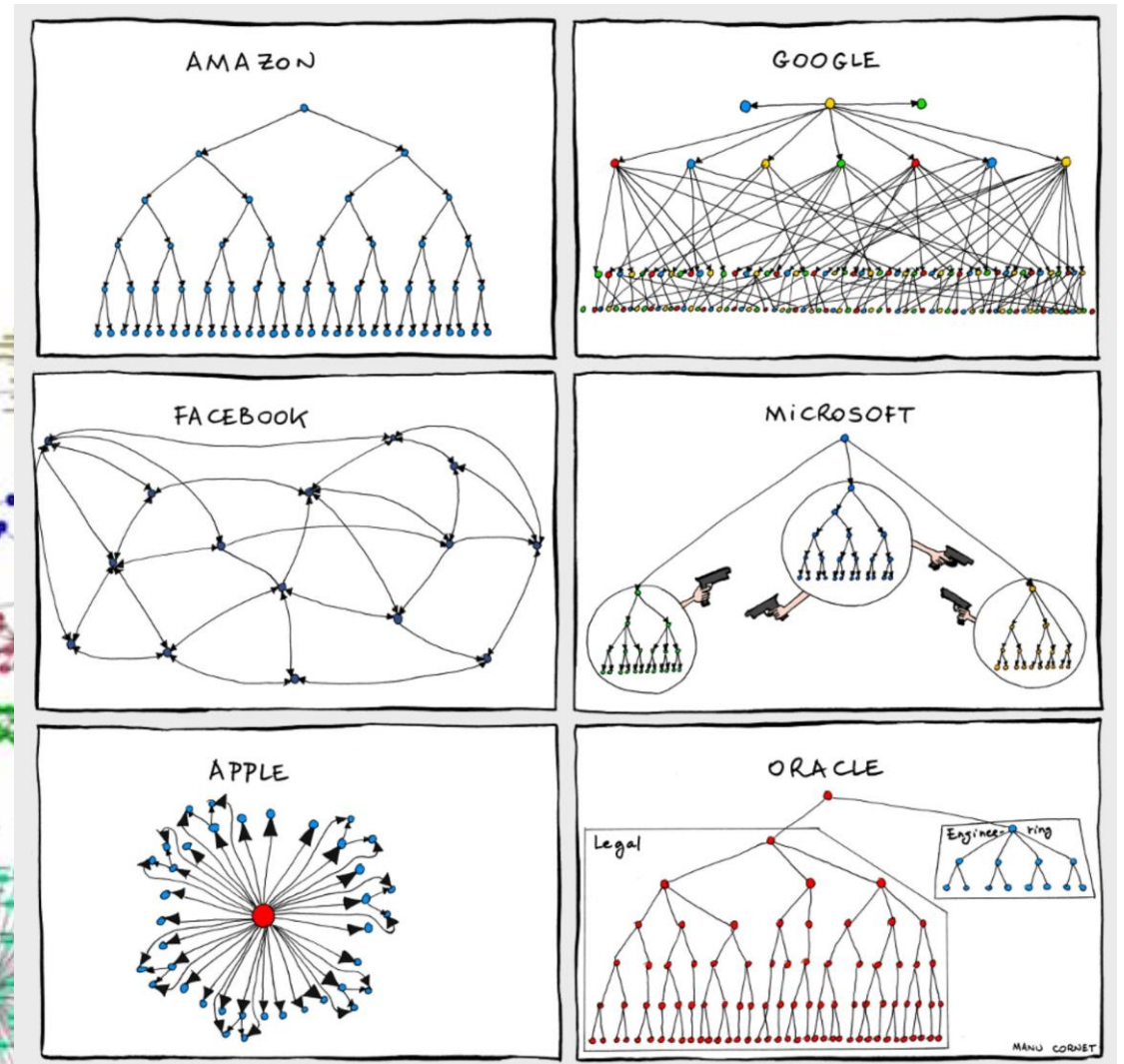
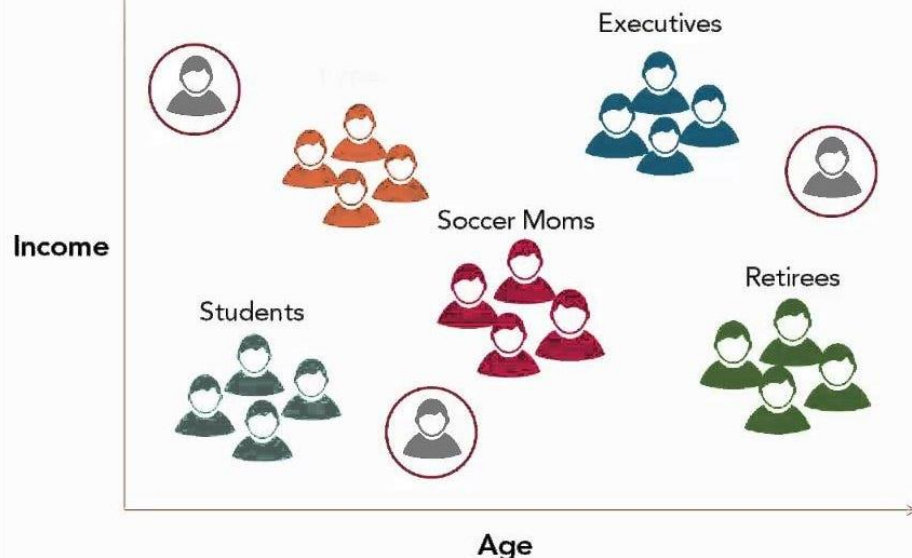
Clustering : k-means clustering

1. Choose the number of clusters K .
2. Initialize **K cluster centroids** randomly
3. Repeat until a maximum number of **iterations**:
 - a. Assign each data point to the nearest **centroid** (based on distance **metric**).
 - b. Update the centroids by computing the **mean** of all their data points assigned
4. Output the final cluster assignments and centroids



Clustering : Applications

- ▶ Social network analysis
- ▶ Anomaly detection (e.g. fraudulent transaction)
- ▶ Market segmentation

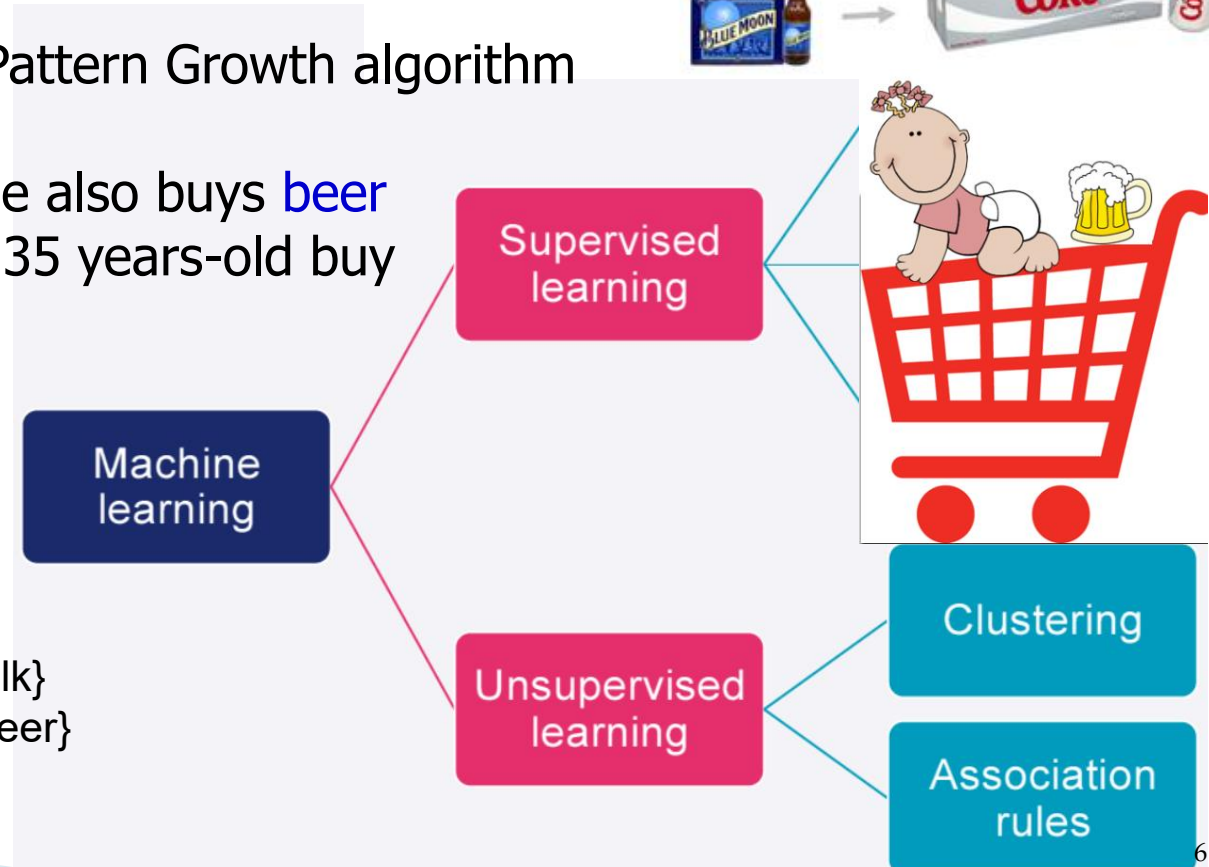


Association Rules

- ▶ Discover **correlation** between any two or more variables
 - **Given:** a set of records containing items
 - **Goal:** Produce **dependency rules**, to predict occurrence of variable X with variable(s) Y
 - **Algorithms:** Apriori algorithm, Frequent Pattern Growth algorithm
 - Market basket analysis
 - If a customer buys **diaper**, very likely he also buys **beer**
 - In the evenings, men between 25 and 35 years-old buy both products
- 
- 

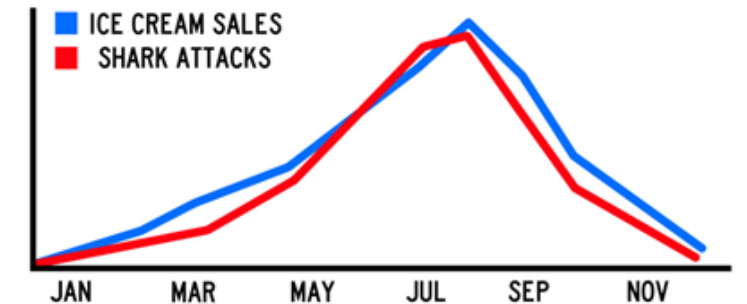
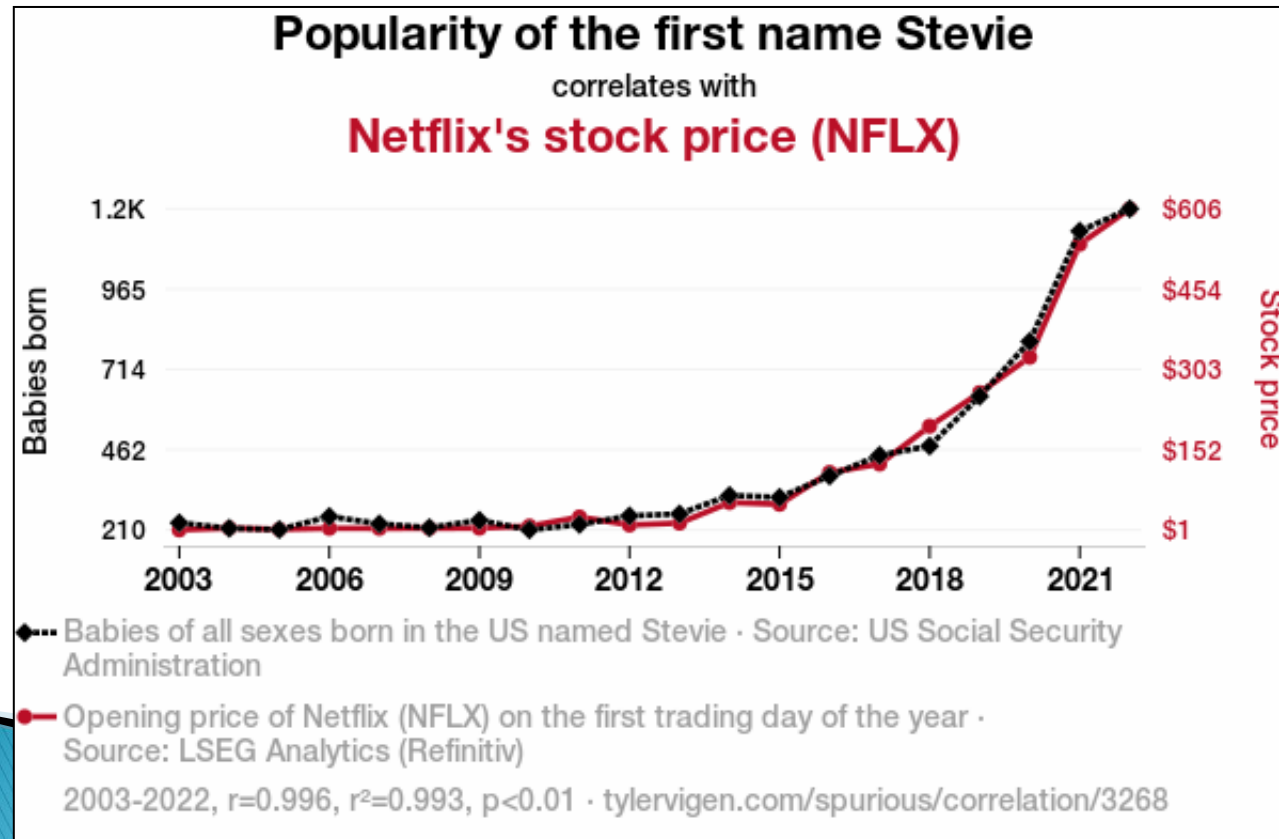
<i>TID</i>	<i>Items</i>
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

{Bread} → {Milk}
{Diaper} → {Beer}

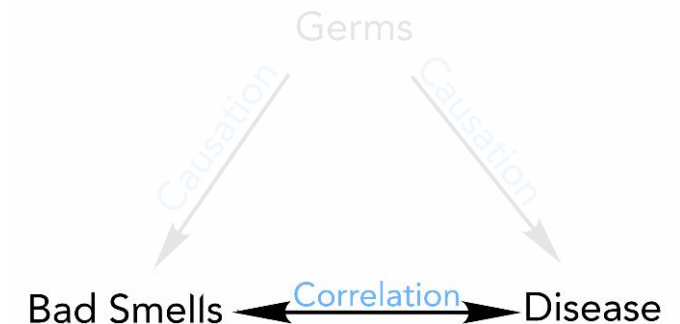


Association (Correlation) vs. Causation

- ▶ Discover **correlation** between any two or more variables
- ▶ **Correlation** $\neq$ **Causation**
- ▶ Data preparation: which input **features**?

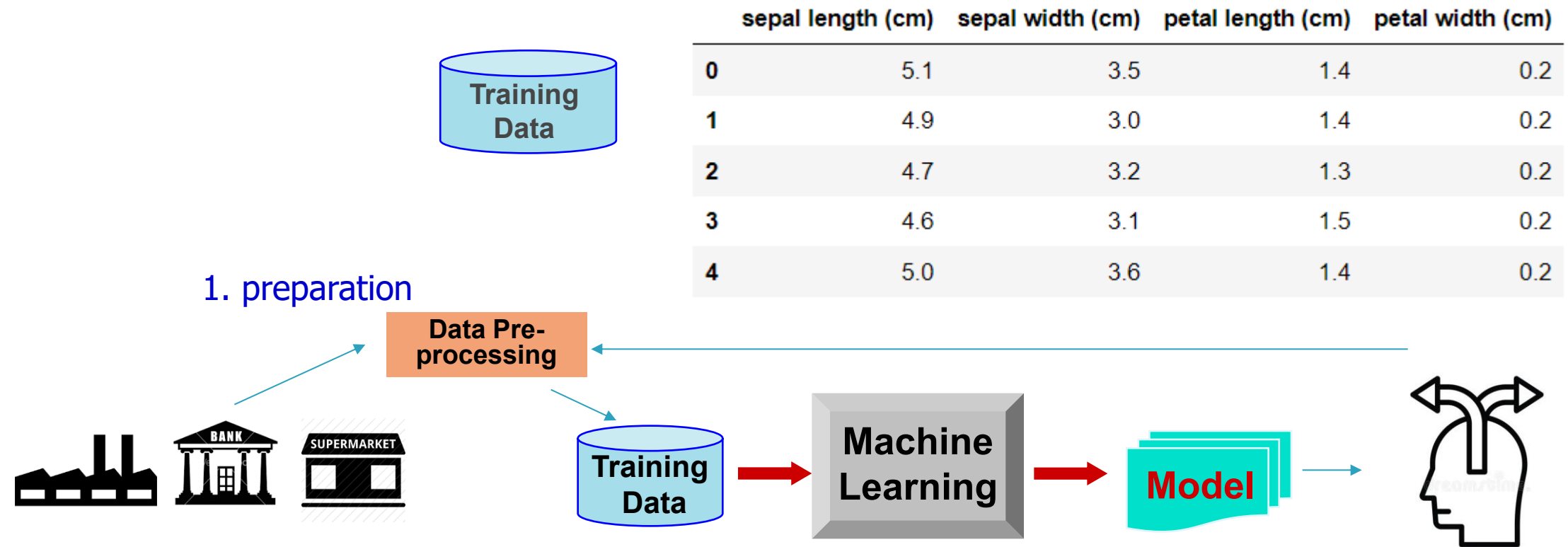


Both ice cream sales and shark attacks increase when the weather is hot and sunny, but they are not caused by each other (they are caused by good weather, with lots of people at the beach, both eating ice cream and having a swim in the sea)



Un-supervised learning: An example

- ▶ Prepare the training



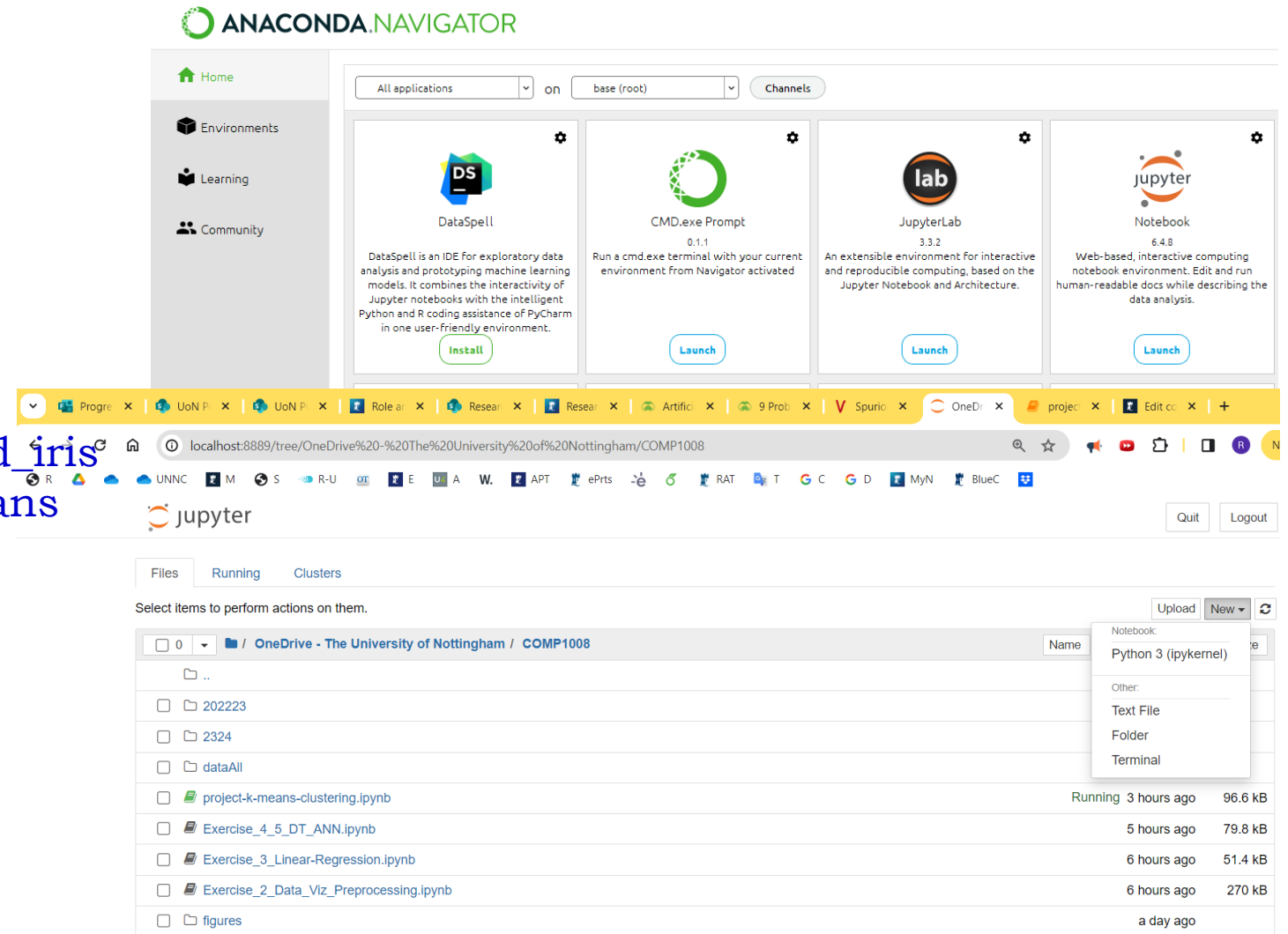
Build a k-means clustering model - implementation

- ▶ Jupyter Notebook

- ▶ Import all libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans
```

Try to build yours in the lab!



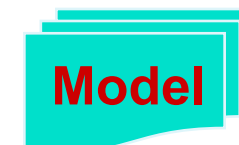
Build a k-means clustering model – data preparation

- ▶ Understand the problem
 - Features
 - Which approaches?

Samples/
Observations/
Instances

	A	B	C	D	E
1	Id	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
2	1	5.1	3.5	1.4	0.2
3	2	4.9	3	1.4	0.2
4	3	4.7	3.2	1.3	0.2
5	4	4.6	3.1	1.5	0.2
6	5	5	3.6	1.4	0.2
7	6	5.4	3.9	1.7	0.4
8	7	4.6	3.4	1.4	0.3
9	8	5	3.4	1.5	0.2
10	9	4.4	2.9	1.4	0.2

Features



Build a k-means clustering model – data preparation

- Read in the **dataset iris**

```
iris = datasets.load_iris()
```

```
#Creating data frame
```

```
Data = pd.DataFrame(iris.data,  
                      columns = iris.feature_names)  
Data.head()
```

	A	B	C	D	E
1	Id	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
2	1	5.1	3.5	1.4	0.2
3	2	4.9	3	1.4	0.2
4	3	4.7	3.2	1.3	0.2
5	4	4.6	3.1	1.5	0.2
6	5	5	3.6	1.4	0.2
	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	
0	5.1	3.5	1.4	0.2	
1	4.9	3.0	1.4	0.2	
2	4.7	3.2	1.3	0.2	
3	4.6	3.1	1.5	0.2	
4	5.0	3.6	1.4	0.2	

1. preparation

Data Pre-
processing



Machine
Learning



Build a k-means clustering model – data preparation

- Observe the dataset iris

Visualising the samples - On the first two columns

```
plt.scatter(Data['sepal length (cm)'], Data['sepal width (cm)'], s = 100)
```

```
plt.xlabel('sepal length')
```

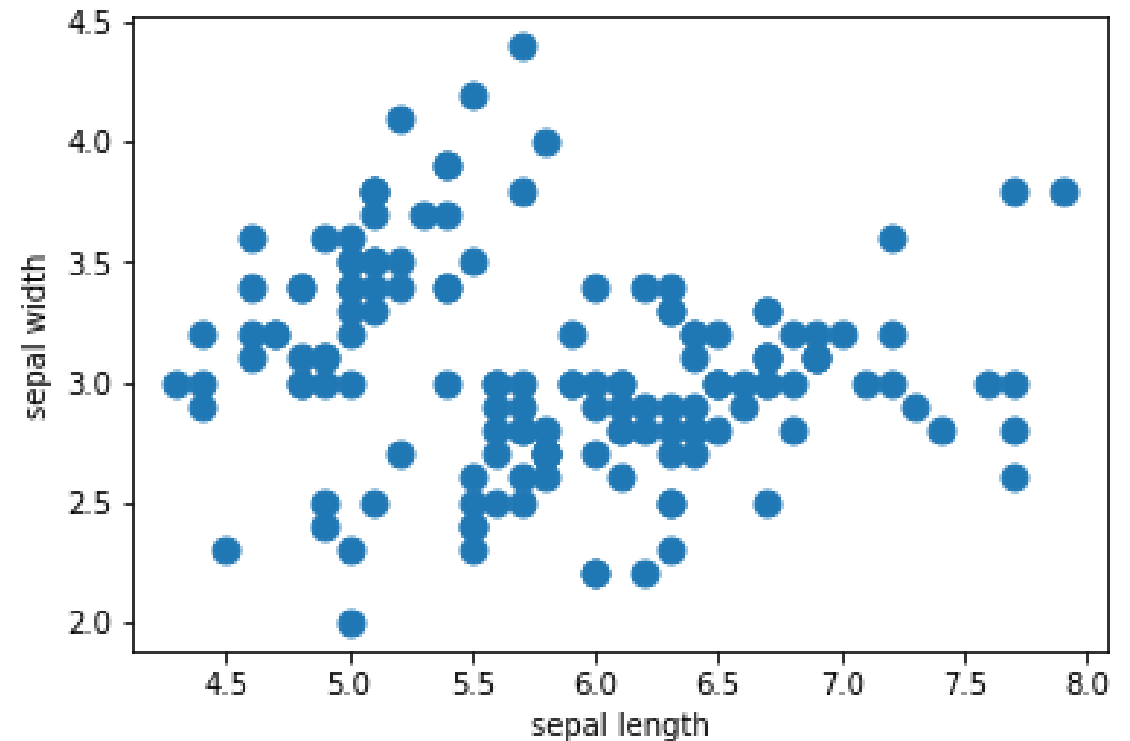
```
plt.ylabel('sepal width')
```

Try yourself:

```
plt.scatter(Data['petal length (cm)'],  
            Data['petal width (cm)'], s = 80)
```

```
plt.xlabel('petal length')
```

```
plt.ylabel('petal width')
```



Build a k-means clustering model – training

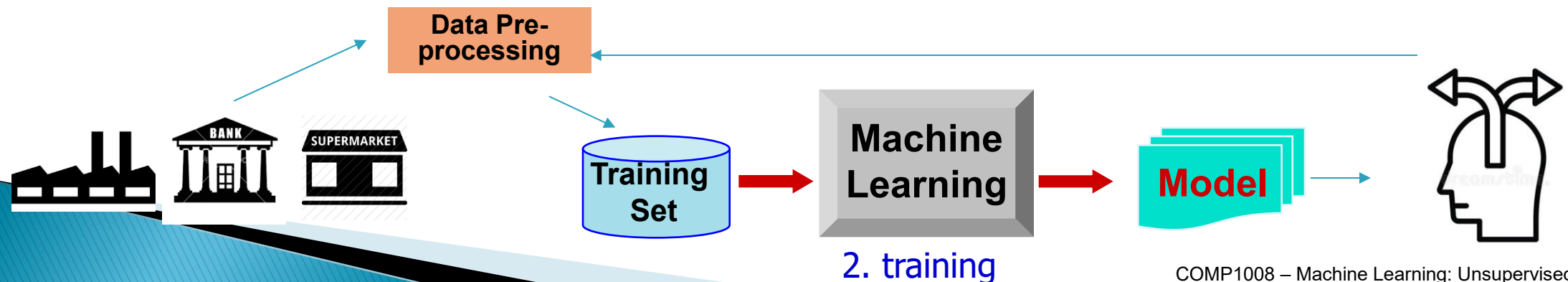
► Training a k-means model

prepare the training data x
`x = Data.values`

create a k-means clustering model
`kmeans = KMeans(n_clusters=3, max_iter = 100, random_state = 0)`

`y_kmeans = kmeans.fit_predict(x)`

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2



Build a k-means clustering model – results

- Where are the **k** centers?

show the centroids

`kmeans.cluster_centers_`

Visualising the clusters - the first two colour

`plt.scatter(x[y_kmeans == 0, 0], x[y_kmeans`

`plt.scatter(x[y_kmeans == 1, 0], x[y_kmeans`

`plt.scatter(x[y_kmeans == 2, 0], x[y_kmeans`

plotting the centroids of the clusters

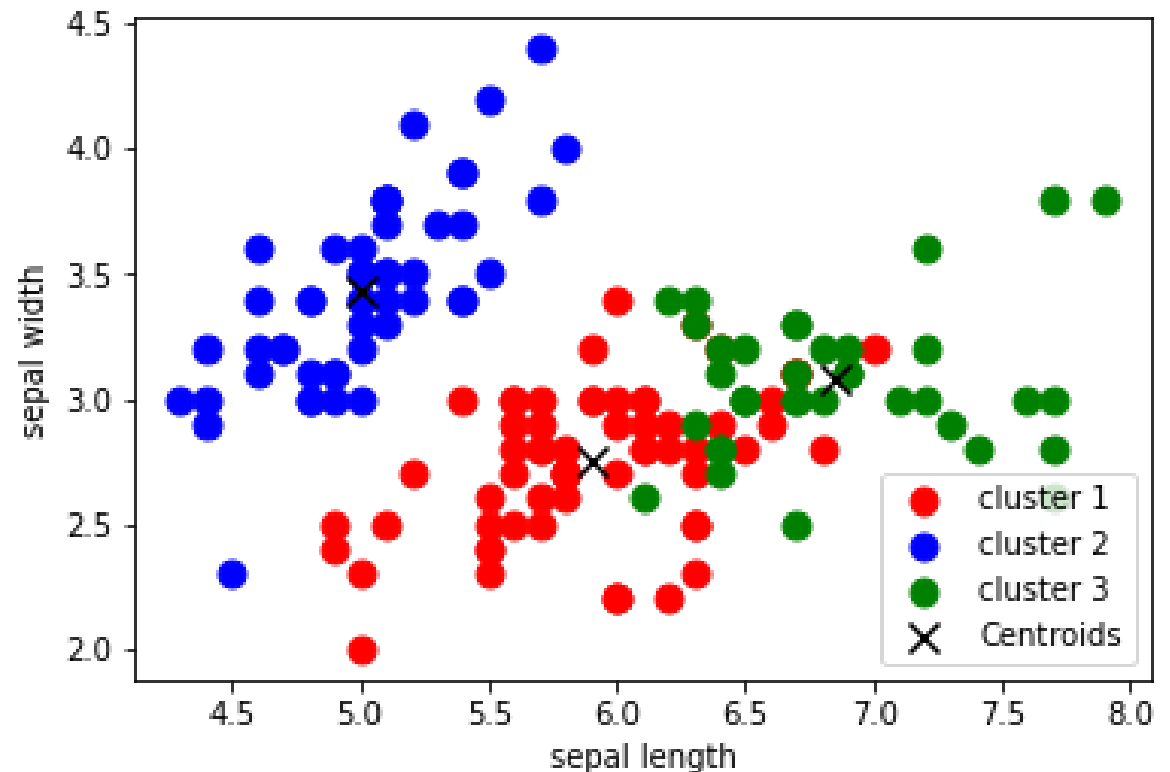
`plt.scatter(kmeans.cluster_centers_[:, 0], km`

`s = 80, c = 'black', label = 'Centroids',`

`plt.legend()`

In [44]: `kmeans.cluster_centers_`

Out[44]: `array([[5.9016129 , 2.7483871 , 4.39354839, 1.43387097],
[5.006 , 3.428 , 1.462 , 0.246],
[6.85 , 3.07368421, 5.74210526, 2.07105263]])`



COMP1008 Part II: Machine Learning

- **Machine learning I:** Intro (last week)
 - Definition
 - Machine learning three pillars
 - **Machine learning II:** [Data & Process](#), Supervised vs. [Un-supervised learning](#) (this week)
 - Supervised vs. un-supervised (difference, process, concepts e.g. features/labels, etc.)
 - Un-supervised learning, k-means, association rules, concepts
 - **Machine learning III:** Supervised learning
 - **Machine learning IV:** ANN
 - Computing session 1: [NumPy and Pandas](#) (this week)
- 